

MobCal-MPI Automation Pipeline

Setup of system for automatic MobCal pipeline

This section details the steps to create an Ubuntu machine which is capable of running the MobCal automation. Steps are provided from the best recollection. Other Linux distributions are likely possible, but development machine was an Ubuntu virtual machine.

Operating System

First the Ubuntu operating system must be downloaded from [Canonical's Website](#). The desktop version was used, but the server version could also be used. Follow the installation instructions provided by Canonical.

If a virtual machine is to be used, then [Oracle VirtualBox](#) must also be downloaded and installed. Virtual machines are handy if the user has a Windows machine and does not want to dual-boot. It also provides finer control over the RAM and Cores used in the system which will be automatically detected by the automation.

Step 1: Get the System ready

Get essentials, compilers, etc

```
$ sudo apt-get update
$ sudo apt-get upgrade
$ sudo apt-get install build-essential
$ sudo apt-get install gfortran
$ sudo apt install bzip2
$ sudo apt-get install cmake
$ sudo apt install git
```

Step 2: Install OpenMPI

Download the OpenMPI version that matches the ORCA binaries (for ORCA 6.0.1 this is OpenMPI 4.1.6) from <https://www.open-mpi.org/software/ompi/v4.1/>, e.g. openmpi-4.1.6.tar.gz. Then [run](#):

```
$ tar -xf openmpi-4.1.6.tar.gz
$ cd openmpi-4.1.6
$ ./configure --prefix="/.openmpi" FC=gfortran
$ make -j 4
$ sudo make install
```

Add to your `.bashrc` file

```
export PATH="$PATH:/.openmpi/bin"
```

```
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:/.openmpi/lib"
```

Additionally, run

```
$ sudo apt install libopenmpi-dev
$ sudo apt install openmpi-bin
```

Check, that mpifort exists, via:

```
$ mpifort --version
```

Step 3: OpenBabel, SDF2TINKERXYZ, and Python

For OpenBabel, get the tar ball from

<https://sourceforge.net/projects/openbabel/files/openbabel/2.4.1/>

Then follow <https://openbabel.org/docs/Installation/install.html#basic-build-procedure>:

```
$ tar -xf openbabel-2.4.1.tar.gz
$ mkdir build
$ cd build
$ cmake ../openbabel-2.4.1
$ make -j 4
$ sudo make install
```

The commands `obabel` and `babel` should be already known to your PATH.

For `sdf2tinkerxyz`, get the tar ball from <https://sourceforge.net/projects/sdf2xyz2sdf/files/source/>

Then run:

```
$ tar xvf sdf2xyz2sdf-1.05.tar.bz2
$ cd sdf2xyz2sdf
$ ./bootstrap
$ ./configure
$ make
$ sudo make install
```

The command `sdf2tinkerxyz` should be already known to your PATH.

For Python, get `python3`. Make sure that `python3` is in your PATH. To get needed packages, run

```
$ sudo apt install python3-pip
```

```
$ pip install numpy scipy matplotlib gitpython pandas
```

Step 4: xTB and CREST

For xTB, get the tar ball from <https://github.com/grimme-lab/xtb/releases>, e.g., xtb-6.7.1-linux-x86_64.tar.xz, and run:

```
$ tar -xf xtb-6.7.1-linux-x86_64.tar.xz
```

Add to your .bashrc file

```
export PATH="$PATH:/home/$USER/xtb-dist/bin"          or
export PATH="$PATH:/opt/xtb-dist/bin"
```

For CREST, get the tar ball from <https://github.com/crest-lab/crest/releases>, e.g., crest-gnu-12-ubuntu-latest.tar.xz (make sure it's the gnu, not the intel one) and run:

```
$ tar -xf crest-gnu-12-ubuntu-latest.tar.xz
```

Add to your .bashrc file

```
export PATH="$PATH:/home/$USER/crest"                or
export PATH="$PATH:/opt/crest"
```

Step 5: ORCA

You need an account in the [ORCA Forum](#) for this. Here, download the latest ORCA version as tar ball (precompiled and linked to specific OpenMPI). Then run:

```
$ tar -xf orca_6_0_1_linux_x86-64_shared_openmpi416.tar.xz
```

Add to your .bashrc file

```
export PATH="$PATH:/home/$USER/orca"
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:/home/$USER/orca"
or:
export PATH="$PATH:/opt/orca"
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:/opt/orca"
```

Note, that despite this, you always want to call ORCA using its full path, especially when running in MPI mode, i.e.,:

```
$ /home/$USER/orca/orca file.inp >& file.out &
```

Step 6: GIT and MobCal

We already installed git in Step 1. Clone the MobCal repository by running

```
$ git clone git@github.com:HopkinsLaboratory/MobCal-MPI\_21.git
```

Currently, this repo is not public so you need a git account for this and be invited. You also need to [create a public ssh key](#)

```
$ ssh-keygen -t ed25519 -C "your_email@example.com"
```

and [add that to your git account](#). Once the repo goes public, this step should be simpler.

Commented [AH1]: Since we are public now, this part is likely different!

Making sure, everything works:

Your `.bashrc` file should now contain

```
export PATH="$PATH:/openmpi/bin"
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:/openmpi/lib"
export PATH="$PATH:/home/$USER/xtb-dist/bin"
export PATH="$PATH:/home/$USER/crest"
export PATH="$PATH:/open3dtools/bin" # not needed if sdf2tinkerxyz
was installed via make install
export PATH="$PATH:/home/$USER/orca"
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:/home/$USER/orca"
```

Using the `which` command, the following commands should be recognized:

```
$ which obabel
$ which babel
$ which sdf2tinkerxyz
$ which xtb
$ which crest
$ which orca
$ which python3
$ which openmpi
$ which mpirun
$ which mpifort
```